

RES-ML — Machine-Learning Methodology

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1 Domain Purpose and Scope

RES-ML defines the machine-learning methodology for the revised research scope. Under the 2026-07-19 scope decision, ML is deferred future work. The active Studentship objective is to stabilise the 2011 Tohoku hydrodynamic pipeline, Kamaishi/Sendai corridor comparison and fixed-rigid barrier-class metrics before any supervised-learning, surrogate optimisation or generative geometry work begins.

The high-fidelity simulation workflow remains the authoritative source of physical response. A future surrogate may be an assessment, screening or sampling tool; it shall not be represented as replacing validated fluid simulation, structural analysis or direct confirmation.

Machine-learning training is blocked until the selected tsunami event and the relevant hydrodynamic models have passed the V1–V2–V3 validation hierarchy recorded in the current scope decision. Historical or reconstructed defence geometries are evidence about plausible cases, not direct causal labels for optimal reinforcement; controlled, validated simulations remain the only acceptable future training targets.

1.1 Active Scope

There is no active ML training scope in the current Research phase. Future RES-ML work may cover:

- response surrogates for validated hydrodynamic outputs;
- optional adequacy or damage classification after structural extensions exist;
- screening within a bounded barrier-class or intervention space;
- active selection of additional validated simulations;
- dimensionality reduction where correlated outputs or histories justify it;
- small-data algorithm assessment and non-ML baselines;
- predictive uncertainty and out-of-domain detection;
- direct-simulation fallback for uncertain or unsupported queries;
- integration with validated regional and local CFD workflows, and later structural/FSI workflows if activated.

1.2 Deferred Scope

The following are deferred to later research and shall not be represented as initial acceptance requirements:

- unrestricted free-form geometry generation;
- surrogate optimisation during the active hydrodynamic phase;

- reinforcement learning;
- unrestricted inverse design;
- claims of cross-event generalisation;
- direct training from before/after geometry alone;
- replacement of high-fidelity confirmation.

1.3 Supervised-Learning Relationship

The intended supervised-learning relationship is:

$$\begin{aligned} &\text{corridor descriptors} + \text{fixed-rigid geometry descriptors} + \text{validated hydrodynamic-condition descriptors} \\ &\rightarrow \text{hydrodynamic response} + \text{predictive uncertainty} \end{aligned} \quad (1.1)$$

Decision role: Equation (1.1) is a future data relationship, not an active implementation requirement. Structural response and damage labels are excluded until RES-STR activates and validates those extensions.

1.4 Research Gates

The ML workstream is governed by the following gates:

1. **G1 — Data-source feasibility.** Source data, CRS/datum, licence and preprocessing provenance are recorded. No ML training is permitted before this gate.
2. **G2 — Corridor definition.** Kamaishi and Sendai corridor geometry is locked. ML remains deferred.
3. **G3 — Regional replay readiness.** The Tohoku regional model and replay-boundary histories are ready for validation. No ML dataset may be accepted before this gate.
4. **G4 — Local hydrodynamic metric extraction.** Fixed-rigid local runs emit complete quality and metric labels. Future dataset design may begin, but model training remains blocked until validation status is known.
5. **G5 — V1–V2–V3 validation.** Solver capability, Tohoku hydrodynamics and local impact/loading evidence are accepted. Future ML training may be reconsidered after this gate.
6. **G6 — Barrier-class demonstration.** Barrier classes are compared with complete provenance. This is not surrogate validation.

2 Domain Deliverables

3 RES-ML-ROL — Machine-Learning Role and Use-Case Definition

3.1 Purpose and Scope

This Deliverable records ML as a deferred future role under the current Tohoku hydrodynamic scope. The active project shall first produce validated fixed-rigid barrier hydrodynamic outputs for the Kamaishi and Sendai corridors.

Deferred possible use cases are:

- a forward response surrogate;
- a future damage or adequacy classifier after structural extensions exist;
- barrier-class or intervention screening and ranking;
- active-learning support for selecting additional direct simulations;
- reduced-order representation of correlated response outputs.

Deferred use cases are:

- generative geometry;
- unrestricted inverse design;
- reinforcement learning;
- cross-event transfer learning.

3.2 Research Questions

3.3 Terminology and Definitions

3.4 Literature Evidence

3.5 Governing Relations and Quantitative Definitions

3.6 Machine-Learning Role

Each retained use case shall record:

- input features;
- prediction or decision target;

-
- required dataset and simulation provenance;
 - non-ML baseline;
 - expected advantage over the baseline;
 - consequence of an incorrect prediction;
 - uncertainty requirement;
 - direct-simulation fallback.

The consequence analysis shall distinguish high-consequence unsafe predictions from conservative but inefficient safe-to-unsafe predictions.

3.7 Non-ML Baseline

3.8 Input and Output Representation

3.9 Dataset Requirements

3.10 Model or Architecture Candidates

3.11 Training and Validation Method

3.12 Evaluation Metrics

3.13 Generalisation and Uncertainty

3.14 Simulation and Optimisation Integration

3.15 Candidate Approaches

3.16 Critical Comparison

3.17 Assumptions and Applicability

3.18 Limitations and Failure Conditions

3.19 Project-Specific Conclusions

The active project has no ML training role. A future surrogate may screen candidate options and identify cases for further simulation, but it shall not autonomously declare a defence safe or replace validated hydrodynamic and structural evidence.

3.20 Selected Approach and Rationale

3.21 Unresolved Decisions

Unresolved questions are the validated output dataset, accepted hydrodynamic labels, future structural/damage labels, and the threshold at which surrogate uncertainty would trigger direct simulation.

3.22 Requirements and Handoffs

3.23 Deliverable Completion Record

4 RES-ML-DAT — Learning Dataset and Simulation-Data Requirements

4.1 Purpose and Scope

This Deliverable records future learning-dataset requirements only. The active project builds hydrodynamic source, corridor, geometry and output manifests; it does not create an ML training dataset. A future dataset sample would be a complete simulation case, not an extracted time step. The deferred dataset relationship is:

$$\begin{aligned} &\text{corridor identity} + \text{fixed-rigid barrier geometry} + \text{validated tsunami condition} + \text{simulation fidelity and num} \\ &\quad \rightarrow \text{hydraulic response} + \text{uncertainty and quality state} \end{aligned} \tag{4.1}$$

Historical pre-event and current or reconstructed geometries may define realistic intervention features and contextual or weak labels, but they shall not be treated as direct causal ground truth for optimal reinforcement.

4.2 Research Questions

4.3 Terminology and Definitions

4.4 Literature Evidence

4.5 Governing Relations and Quantitative Definitions

4.6 Machine-Learning Role

4.7 Non-ML Baseline

4.8 Input and Output Representation

4.9 Dataset Requirements

Future ML data groups may include:

- corridor, geometry and case identity;
- fixed-rigid barrier geometry;
- local tsunami-condition descriptors;
- fluid-response outputs;
- simulation metadata;
- convergence state;
- physical-validity state;
- uncertainty state;
- future structural, damage or adequacy labels only if those extensions are activated and validated.

No ML training dataset may be accepted before the selected event, local hydrodynamic model and impact/loading outputs have passed the V1–V2–V3 validation gates.

4.10 Model or Architecture Candidates

4.11 Training and Validation Method

The previous mandatory secondary-event holdout requirement remains removed from the active project. Future ML validation would require grouped partitioning based on complete simulation cases, held-out geometry configurations, held-out condition ranges and independent final test simulations.

Time steps from the same simulation shall never be randomly split between future training and test datasets.

4.12 Evaluation Metrics

4.13 Generalisation and Uncertainty

4.14 Simulation and Optimisation Integration

4.15 Candidate Approaches

4.16 Critical Comparison

4.17 Assumptions and Applicability

4.18 Limitations and Failure Conditions

4.19 Project-Specific Conclusions

The initial learning dataset is a controlled simulation-derived dataset for one validated event-conditioned domain. It is not a broad multi-event dataset and does not support cross-event generalisation claims.

4.20 Selected Approach and Rationale

4.21 Unresolved Decisions

4.22 Requirements and Handoffs

4.23 Deliverable Completion Record

5 RES-ML-REP — Input, Output and Geometry Representation

5.1 Purpose and Scope

This Deliverable records future ML representation requirements. The active representation work is the RES-GEO/RES-DAT manifest for fixed- rigid barrier-class hydrodynamic comparison. Raw voxel, mesh, graph, full-field learning and learned geometry representations are deferred.

5.2 Research Questions

5.3 Terminology and Definitions

5.4 Literature Evidence

5.5 Governing Relations and Quantitative Definitions

5.6 Machine-Learning Role

5.7 Non-ML Baseline

5.8 Input and Output Representation

Future input descriptors may include corridor identity, barrier class, height, width and thickness, orientation, face angle, resolved openings, incident water depth, incident velocity, wave direction, duration or time-history descriptors. Material, foundation and reinforcement descriptors are excluded until structural or optimisation extensions are activated.

Future outputs may include peak force, peak overturning moment, overtopping, run-up and energy-transfer or dissipation metrics. Displacement, stress, strain, permanent deformation, adequacy class, damage class and structural margin are deferred.

5.9 Dataset Requirements

5.10 Model or Architecture Candidates

5.11 Training and Validation Method

5.12 Evaluation Metrics

5.13 Generalisation and Uncertainty

Every representation shall carry feature provenance and known uncertainty so that missing geometry, uncertain material properties and inferred foundation conditions are not silently treated as measured quantities.

5.14 Simulation and Optimisation Integration

5.15 Candidate Approaches

5.16 Critical Comparison

5.17 Assumptions and Applicability

5.18 Limitations and Failure Conditions

5.19 Project-Specific Conclusions

The current representation family is a physics-informed manifest for fixed-rigid hydrodynamic comparison. It may later become a tabular ML descriptor set, but it is not currently a surrogate input, generative geometry encoding or unrestricted design-space representation.

5.20 Selected Approach and Rationale

5.21 Unresolved Decisions

5.22 Requirements and Handoffs

5.23 Deliverable Completion Record

6 RES-ML-ALG — Learning Algorithm and Architecture Assessment

6.1 Purpose and Scope

This Deliverable records future algorithm candidates only. The active project does not assess or select an ML algorithm; it first produces validated hydrodynamic outputs and structured provenance.

6.2 Research Questions

6.3 Terminology and Definitions

6.4 Literature Evidence

The Intelligent Systems lecture material may be used as terminology and foundation context for linear regression, polynomial regression, logistic regression, Naive Bayes, model selection, kernels, support-vector methods, clustering, dimensionality reduction, neural networks, linear

algebra, calculus, probability and statistics. It shall not be used as the sole evidence for research claims that require peer-reviewed or authoritative primary sources.

6.5 Governing Relations and Quantitative Definitions

6.6 Machine-Learning Role

6.7 Non-ML Baseline

6.8 Input and Output Representation

6.9 Dataset Requirements

6.10 Model or Architecture Candidates

If ML is reactivated later, the assessment hierarchy may be:

1. linear regression baseline;
2. regularised polynomial regression;
3. logistic regression for classification;
4. linear and nonlinear support-vector methods;
5. kernel-based regression;
6. tree ensembles or boosting;
7. Gaussian-process methods where computationally feasible;
8. feed-forward neural networks only when dataset volume supports them;
9. PCA or another controlled dimensionality-reduction method for correlated outputs or histories;
10. clustering for exploratory analysis only.

6.11 Training and Validation Method

6.12 Evaluation Metrics

6.13 Generalisation and Uncertainty

6.14 Simulation and Optimisation Integration

6.15 Candidate Approaches

6.16 Critical Comparison

Model selection shall consider dataset size, dimensionality, nonlinearity, uncertainty, interpretability, extrapolation, failure-boundary behaviour and implementation cost. Popularity or implementation convenience alone is insufficient evidence for selection.

6.17 Assumptions and Applicability

6.18 Limitations and Failure Conditions

6.19 Project-Specific Conclusions

The initial algorithm strategy is baseline-first and uncertainty-aware. A final algorithm remains unresolved until data volume, feature dimensionality and validation behaviour are known.

6.20 Selected Approach and Rationale

6.21 Unresolved Decisions

6.22 Requirements and Handoffs

6.23 Deliverable Completion Record

7 RES-ML-OBJ — Loss Functions, Rewards and Physical Constraints

7.1 Purpose and Scope

This Deliverable defines prediction losses, classification labels and engineering constraints for the event-conditioned supervised-learning task. Reinforcement-learning rewards are removed from the active scope.

7.2 Research Questions

7.3 Terminology and Definitions

7.4 Literature Evidence

7.5 Governing Relations and Quantitative Definitions

7.6 Machine-Learning Role

7.7 Non-ML Baseline

7.8 Input and Output Representation

7.9 Dataset Requirements

7.10 Model or Architecture Candidates

7.11 Training and Validation Method

7.12 Evaluation Metrics

The evaluation methodology shall include regression losses, classification losses, multi-output normalisation, regularisation, class imbalance handling, physical bounds, invalid prediction handling, and asymmetric consequences of classification errors.

The following error asymmetry is mandatory:

- unsafe defence predicted safe: high-consequence error;
- safe defence predicted unsafe: conservative but inefficient error.

Acceptance criteria and model-selection metrics shall account for this asymmetry rather than relying only on aggregate accuracy.

7.13 Generalisation and Uncertainty

7.14 Simulation and Optimisation Integration

Losses and labels shall support reinforcement screening only within the validated event-conditioned domain. Queries outside the domain or with invalid physical predictions shall be rejected or routed to direct simulation.

7.15 Candidate Approaches

7.16 Critical Comparison

7.17 Assumptions and Applicability

7.18 Limitations and Failure Conditions

7.19 Project-Specific Conclusions

The project uses supervised response and adequacy targets, not reinforcement-learning reward optimisation. Safety-relevant false-safe predictions dominate the risk model.

7.20 Selected Approach and Rationale

7.21 Unresolved Decisions

7.22 Requirements and Handoffs

7.23 Deliverable Completion Record

8 RES-ML-SUR — Surrogate, Multifidelity and Active-Learning Strategy

8.1 Purpose and Scope

This Deliverable records surrogate, multifidelity and active-learning ideas as deferred future work. The active project does not train a surrogate or run active sampling; it prepares validated hydrodynamic outputs for possible future use.

8.2 Research Questions

8.3 Terminology and Definitions

8.4 Literature Evidence

8.5 Governing Relations and Quantitative Definitions

8.6 Machine-Learning Role

The deferred future query is:

$$\begin{aligned} &\text{corridor} + \text{validated-event condition} + \text{fixed-rigid barrier class} \\ &\rightarrow \text{predicted response} + \text{predictive uncertainty} \end{aligned} \tag{8.1}$$

Different tsunami conditions initially mean controlled variations within a defensible envelope around the validated event, such as local wave-height variation, inundation-depth variation, velocity variation, direction variation and selected severity scaling. This shall not be described as generalisation to unrelated earthquakes, coastlines or tsunami events.

8.7 Non-ML Baseline

8.8 Input and Output Representation

8.9 Dataset Requirements

8.10 Model or Architecture Candidates

8.11 Training and Validation Method

8.12 Evaluation Metrics

8.13 Generalisation and Uncertainty

8.14 Simulation and Optimisation Integration

A future active-sampling strategy may prioritise high-uncertainty hydrodynamic regions, underrepresented geometry configurations and cases where model disagreement is high. Failure-boundary and direct FSI confirmation cases remain deferred until structural/FSI extensions exist.

8.15 Candidate Approaches

8.16 Critical Comparison

8.17 Assumptions and Applicability

8.18 Limitations and Failure Conditions

8.19 Project-Specific Conclusions

A future surrogate may be an event-conditioned screening tool. Direct simulation remains the source of accepted physical response and is required whenever uncertainty, domain limits or consequence level demand confirmation.

8.20 Selected Approach and Rationale

8.21 Unresolved Decisions

8.22 Requirements and Handoffs

8.23 Deliverable Completion Record

9 RES-ML-TRN — Training, Validation and Experiment Methodology

9.1 Purpose and Scope

This Deliverable records future training, grouped validation and model-selection rules only. There is no active ML training in the current hydrodynamic Research phase. If ML is reactivated later, the simulation run shall be the indivisible grouping unit.

9.2 Research Questions

9.3 Terminology and Definitions

9.4 Literature Evidence

9.5 Governing Relations and Quantitative Definitions

9.6 Machine-Learning Role

9.7 Non-ML Baseline

9.8 Input and Output Representation

9.9 Dataset Requirements

9.10 Model or Architecture Candidates

9.11 Training and Validation Method

If ML is reactivated later, the previous secondary-event holdout requirement shall be replaced with:

- grouped splits by complete simulation;
- held-out geometry or intervention options;
- held-out condition ranges;
- structure-level holdout where sufficient structures exist;
- grouped or nested cross-validation;
- fixed seeds;
- non-ML baselines;
- independent final test cases.

Time steps or spatial samples from one simulation shall not be randomly divided between future training and test partitions.

9.12 Evaluation Metrics

9.13 Generalisation and Uncertainty

9.14 Simulation and Optimisation Integration

9.15 Candidate Approaches

9.16 Critical Comparison

9.17 Assumptions and Applicability

9.18 Limitations and Failure Conditions

9.19 Project-Specific Conclusions

Future ML validation shall be grouped by engineering case and geometry/intervention option, not by arbitrary sample rows. This protects against leakage from repeated outputs of the same simulation campaign.

9.20 Selected Approach and Rationale

9.21 Unresolved Decisions

9.22 Requirements and Handoffs

9.23 Deliverable Completion Record

10 RES-ML-GEN — Generalisation, Uncertainty and Interpretability

10.1 Purpose and Scope

This Deliverable restricts generalisation claims to the defined event-conditioned domain. Cross-event performance is removed as an initial acceptance requirement.

10.2 Research Questions

10.3 Terminology and Definitions

10.4 Literature Evidence

10.5 Governing Relations and Quantitative Definitions

10.6 Machine-Learning Role

10.7 Non-ML Baseline

10.8 Input and Output Representation

10.9 Dataset Requirements

10.10 Model or Architecture Candidates

10.11 Training and Validation Method

10.12 Evaluation Metrics

10.13 Generalisation and Uncertainty

The methodology shall define interpolation limits, extrapolation detection, held-out intervention tests, held-out severity tests, uncertainty calibration, out-of-distribution rejection, feature sensitivity, engineering interpretability, permitted-use boundaries, and conditions requiring a new FSI simulation.

10.14 Simulation and Optimisation Integration

10.15 Candidate Approaches

10.16 Critical Comparison

10.17 Assumptions and Applicability

10.18 Limitations and Failure Conditions

The surrogate ceases to be applicable outside the validated event envelope, outside represented defence and reinforcement types, where feature provenance is insufficient, where predictive uncertainty is excessive, or where the engineering consequence of an incorrect prediction requires direct simulation.

10.19 Project-Specific Conclusions

The initial project may claim only within-domain event-conditioned performance. Cross-event generalisation remains future work.

10.20 Selected Approach and Rationale

10.21 Unresolved Decisions

10.22 Requirements and Handoffs

10.23 Deliverable Completion Record

11 RES-ML-INT — Integration with Simulation and Optimisation

11.1 Purpose and Scope

This Deliverable records a deferred integration path. During the active project phase, Software integration is limited to data handling, structured outputs and validation reports for the regional and local hydrodynamic models. Surrogate, optimisation and FSI integration remain future work.

- 11.2 Research Questions**
- 11.3 Terminology and Definitions**
- 11.4 Literature Evidence**
- 11.5 Governing Relations and Quantitative Definitions**
- 11.6 Machine-Learning Role**
- 11.7 Non-ML Baseline**
- 11.8 Input and Output Representation**
- 11.9 Dataset Requirements**
- 11.10 Model or Architecture Candidates**
- 11.11 Training and Validation Method**
- 11.12 Evaluation Metrics**
- 11.13 Generalisation and Uncertainty**
- 11.14 Simulation and Optimisation Integration**

The deferred integration sequence is:

1. consume validated Tohoku corridor and barrier-class outputs;
2. specify a future candidate configuration or intervention;
3. verify that the query lies inside the future surrogate domain;
4. predict response and uncertainty;
5. screen or rank reinforcement options;
6. require direct hydrodynamic, structural or FSI simulation when uncertain or out of domain;
7. add accepted verified simulations to the controlled dataset.

11.15 Candidate Approaches

11.16 Critical Comparison

11.17 Assumptions and Applicability

11.18 Limitations and Failure Conditions

11.19 Project-Specific Conclusions

Integration remains a governed future screening workflow. For the current phase, accepted physical evidence flows from validated simulation into structured outputs only; surrogate predictions do not route decisions.

11.20 Selected Approach and Rationale

11.21 Unresolved Decisions

11.22 Requirements and Handoffs

11.23 Deliverable Completion Record

12 RES-ML-SPEC — Integrated Machine-Learning Specification

12.1 Purpose and Scope

This Deliverable records that the event-conditioned defence-assessment ML specification is deferred under the current hydrodynamic scope. It preserves the WBS ID without creating a replacement hierarchy.

12.2 Research Questions

12.3 Terminology and Definitions

12.4 Literature Evidence

12.5 Governing Relations and Quantitative Definitions

12.6 Machine-Learning Role

Any future ML specification shall consolidate candidate ML roles, historical evidence, simulation-derived data, barrier and intervention representations, algorithm hierarchy, losses and labels, grouped validation, uncertainty, out-of-domain handling, simulation fallback, reinforcement-screening workflow, limitations and extension gates.

12.7 Non-ML Baseline

12.8 Input and Output Representation

12.9 Dataset Requirements

12.10 Model or Architecture Candidates

12.11 Training and Validation Method

12.12 Evaluation Metrics

12.13 Generalisation and Uncertainty

12.14 Simulation and Optimisation Integration

12.15 Candidate Approaches

12.16 Critical Comparison

12.17 Assumptions and Applicability

12.18 Limitations and Failure Conditions

12.19 Project-Specific Conclusions

12.20 Selected Approach and Rationale

The selected active approach is no ML implementation or training. Future event-conditioned supervised learning may be reconsidered only after the Tohoku hydrodynamic workflow, fixed-rigid barrier metrics and V1–V2–V3 validation evidence are accepted.

12.21 Unresolved Decisions

12.22 Requirements and Handoffs

The integrated specification passes the following requirements downstream:

- no ML training before event, local hydrodynamic and impact/loading validation gates;
- no claim that historical geometry changes are optimal causal labels;
- grouped validation by complete simulation case;
- explicit uncertainty and out-of-domain rejection;

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- direct hydrodynamic, structural or FSI fallback for uncertain or unsupported future queries;
 - deferred Jira WBS updates through the pending change manifest.

12.23 Deliverable Completion Record